Kubeflow作業與工作範例

NetApp Solutions

NetApp
May 03, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/netapp-solutions/ai/aicp_example_kubeflow_operations_and_tasks_overview.html on May 03, 2022. Always check docs.netapp.com for the latest.

目錄

Kubeflow作業與工作範例 1
Kubeflow作業與工作範例 1
為資料科學家或開發人員提供Jupyter筆記型電腦工作區 1
筆記型電腦和管線範例 9

Kubeflow作業與工作範例

本節包含使用Kubeflow執行的各種作業和工作範例。

"下一步：為資料科學家或開發人員提供Jupyter筆記型電腦工作區。"

Kubeflow作業與工作範例

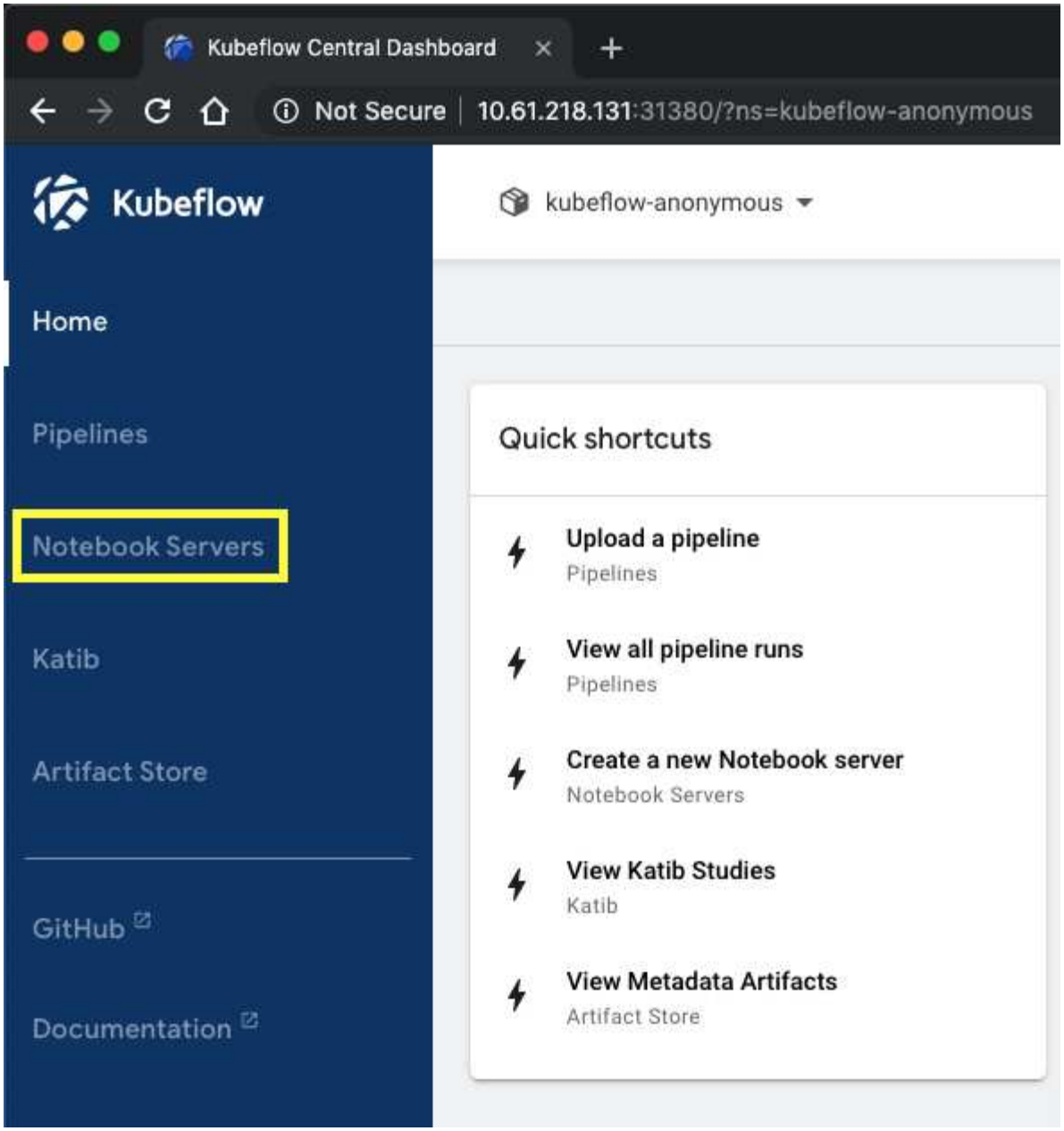
本節包含使用Kubeflow執行的各種作業和工作範例。

"下一步：為資料科學家或開發人員提供Jupyter筆記型電腦工作區。"

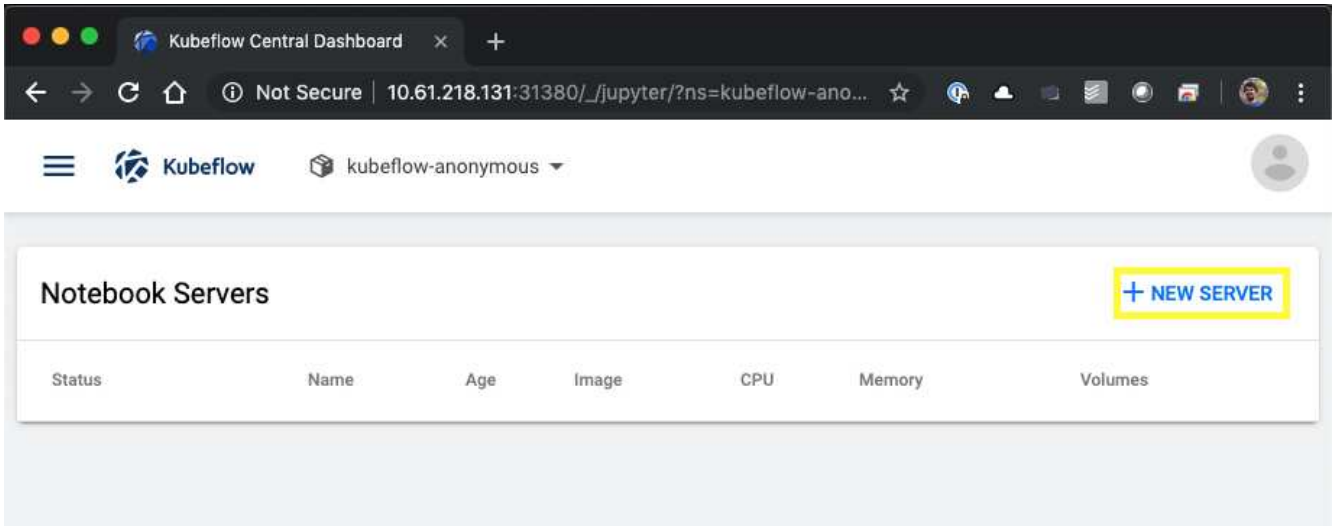
為資料科學家或開發人員提供Jupyter筆記型電腦工作區

Kubeflow能夠快速配置新的Jupyter筆記型電腦伺服器、做為資料科學家工作空間。若要使用Kubeflow來配置新的Jupyter筆記型電腦伺服器、請執行下列工作。如需Kubeflow內容中Jupyter Notebooks的詳細資訊、請參閱 "官方Kubeflow文件"。

1. 在Kubeflow中央儀表板中、按一下主功能表中的「Notebook Servers（筆記型電腦伺服器）」、瀏覽至Jupyter Notebook server管理頁面。

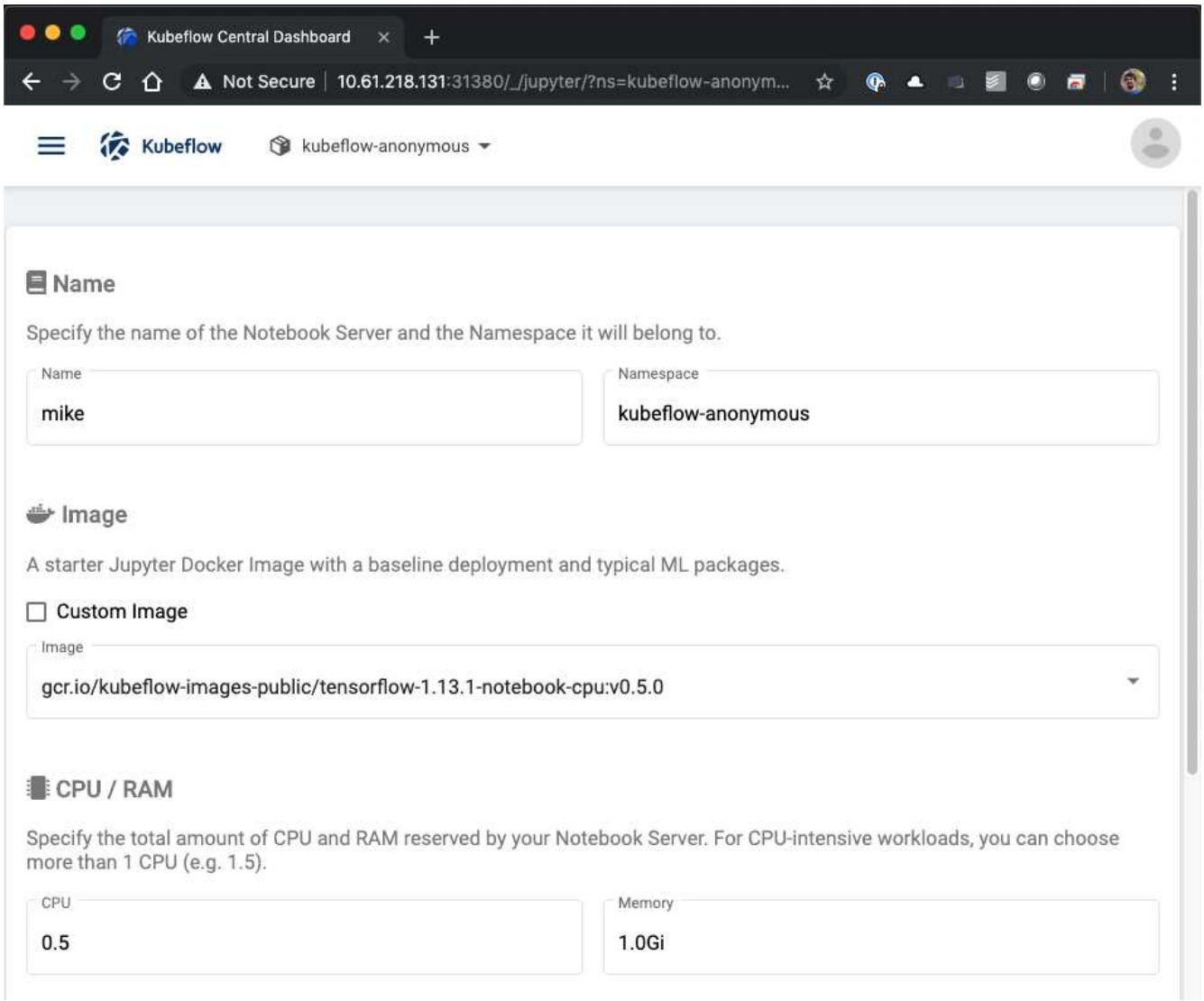


2. 按一下「New Server（新伺服器）」以配置新的Jupyter Notebook伺服器。

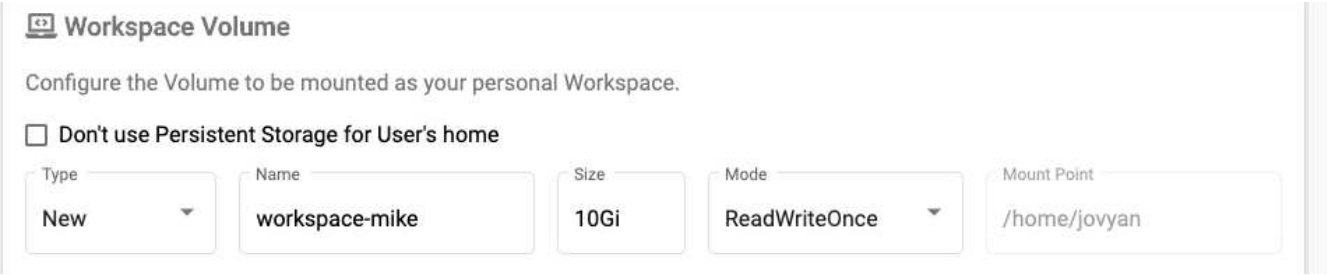


3. 命名新伺服器、選擇您要伺服器所依據的Docker映像、然後指定伺服器要保留的CPU和RAM數量。如果「命名空間」欄位為空白、請使用頁面標題中的「選取命名空間」功能表來選擇命名空間。然後、「Namespace（命名空間）」欄位會自動填入所選的命名空間。

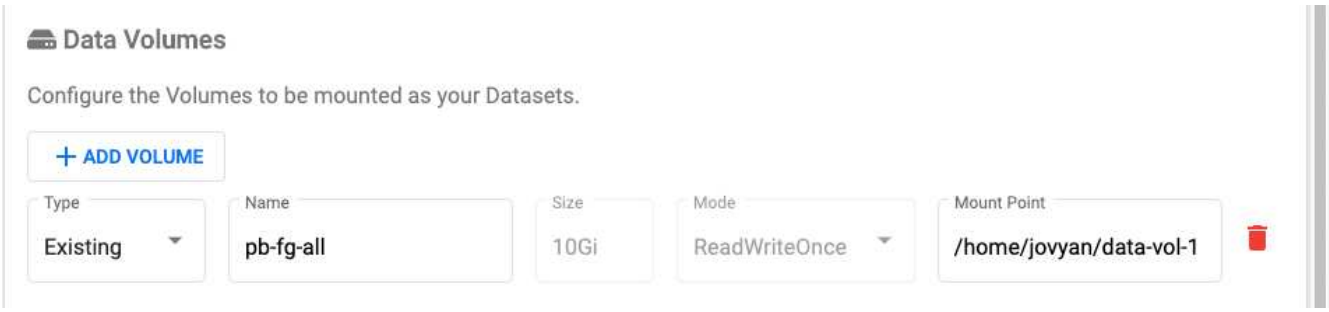
   在以下範例中、會選擇「kubeflow匿名」命名空間。此外、Docker映像、CPU和RAM的預設值也會被接受。



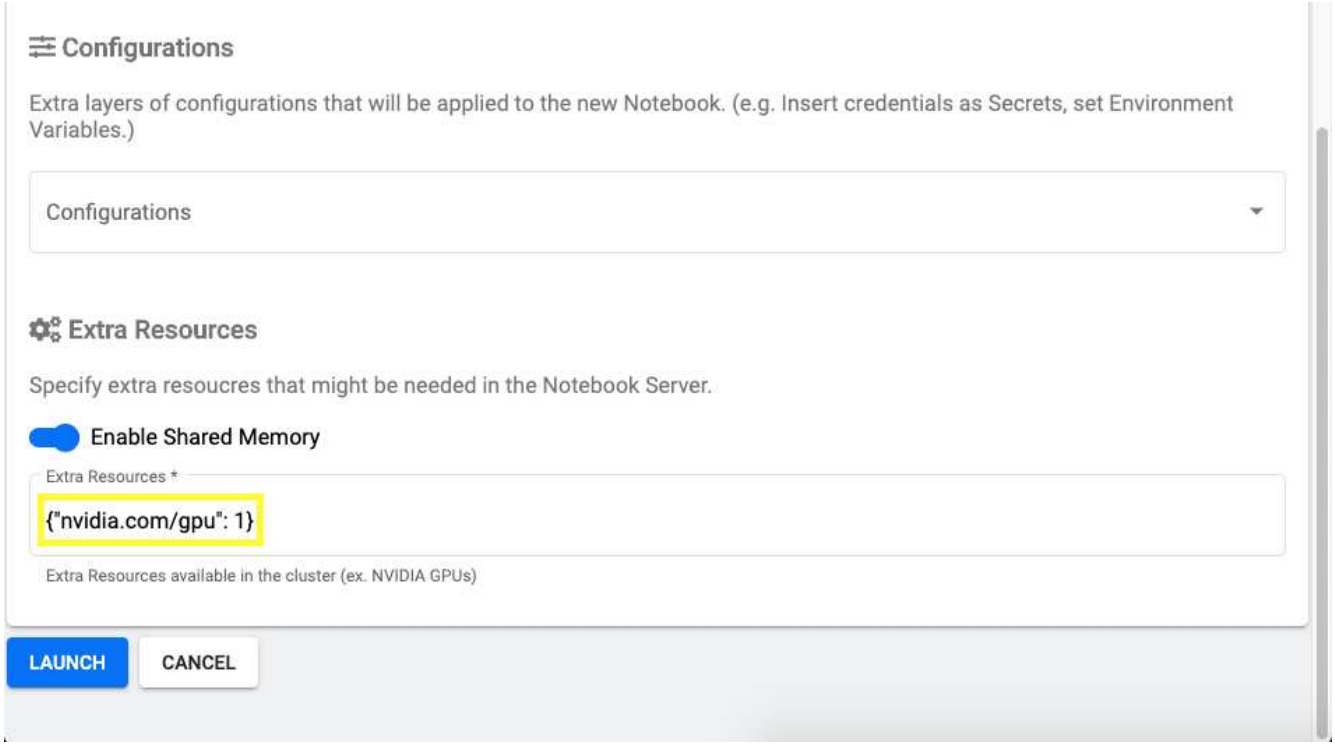
4. 指定工作區Volume詳細資料。如果您選擇建立新磁碟區、則該磁碟區或永久虛擬磁碟區會使用預設StorageClass進行資源配置。因為使用Trident的StorageClass已在區段中指定為預設StorageClass "Kubeflow部署"、磁碟區或永久虛擬磁碟區是以Trident來配置。此Volume會自動掛載為Jupyter Notebook Server Container中的預設工作區。使用者在伺服器上建立的任何筆記型電腦、若未儲存至個別的資料Volume、都會自動儲存至此工作區磁碟區。因此、筆記型電腦會在重新開機後持續運作。



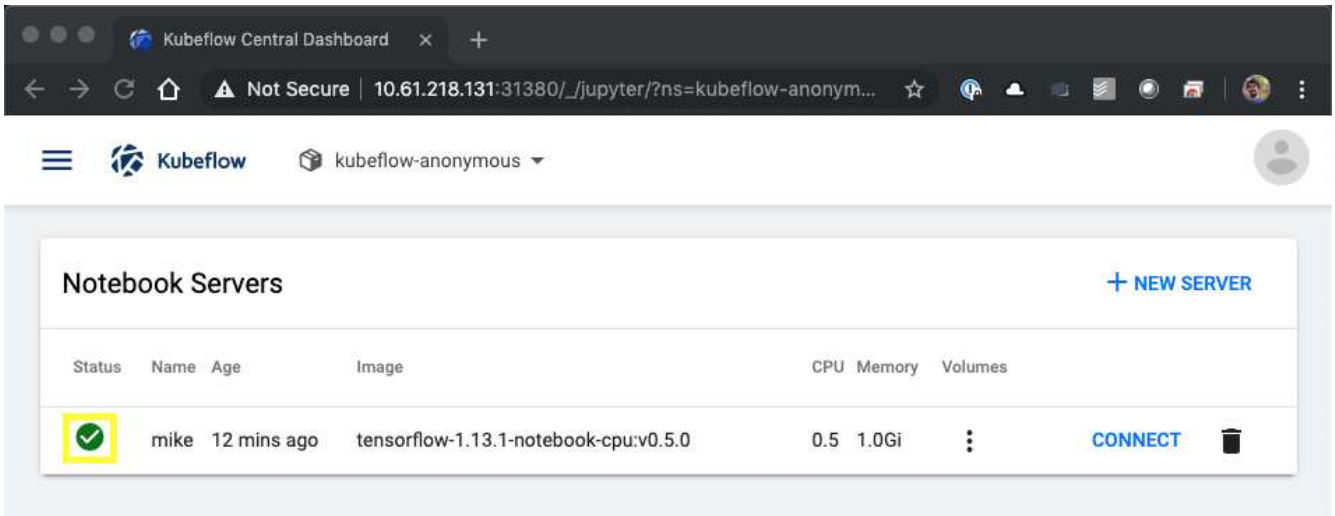
5. 新增資料磁碟區。下列範例指定名為「PPB -FG All」的現有PVc、並接受預設掛載點。



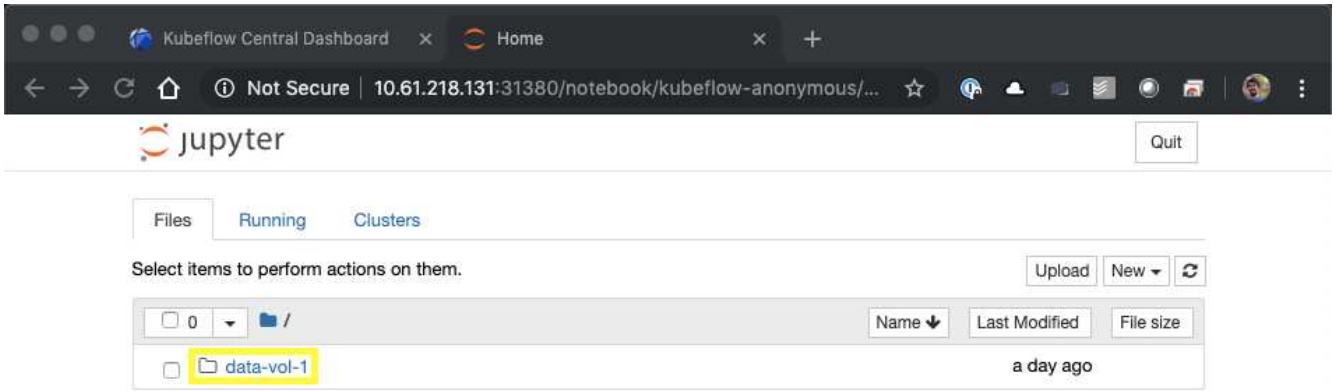
6. *選用：*要求將所需的GPU數量分配給您的筆記型電腦伺服器。在下列範例中、需要一個GPU。

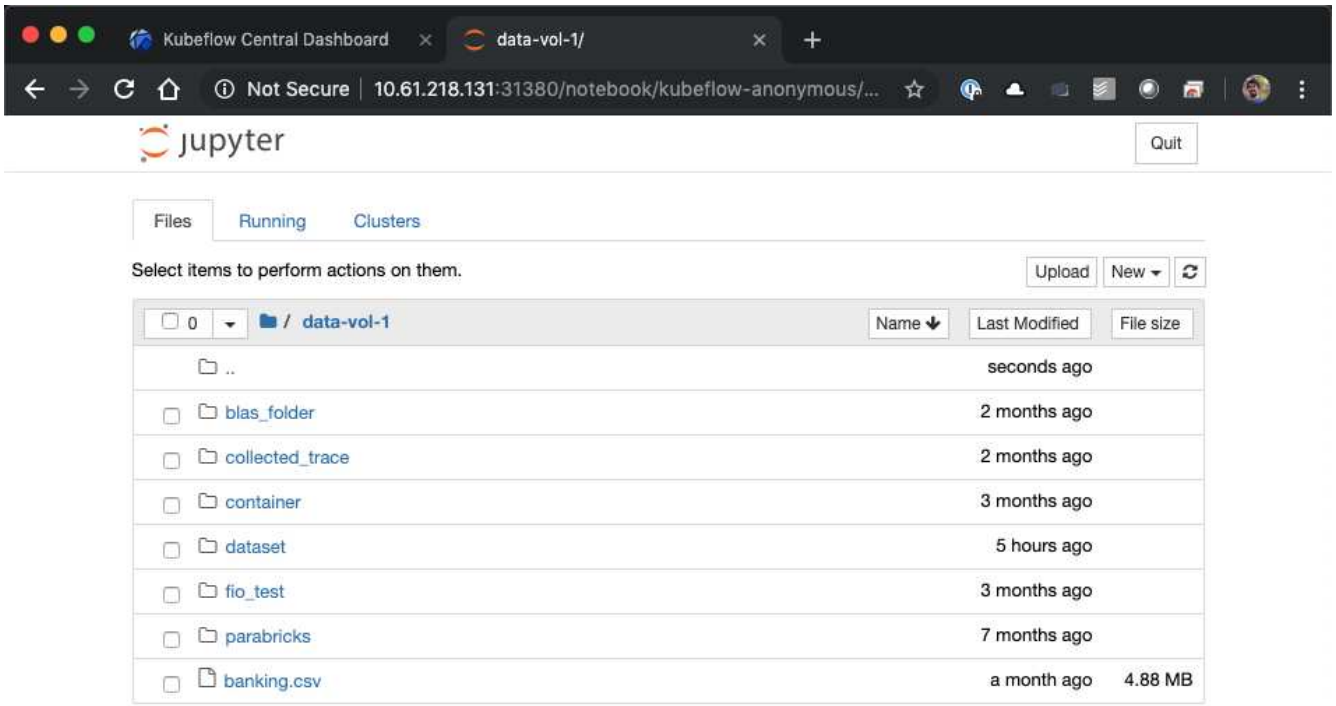


7. 按一下「啟動」以配置新的筆記型電腦伺服器。
8. 等待筆記型電腦伺服器完全配置完成。如果您從未使用您指定的Docker映像來配置伺服器、則可能需要幾分鐘的時間、因為該映像需要下載。當伺服器已完全配置完成時、您會在Jupyter筆記型電腦管理頁面的「Status（狀態）」欄中看到綠色勾號。



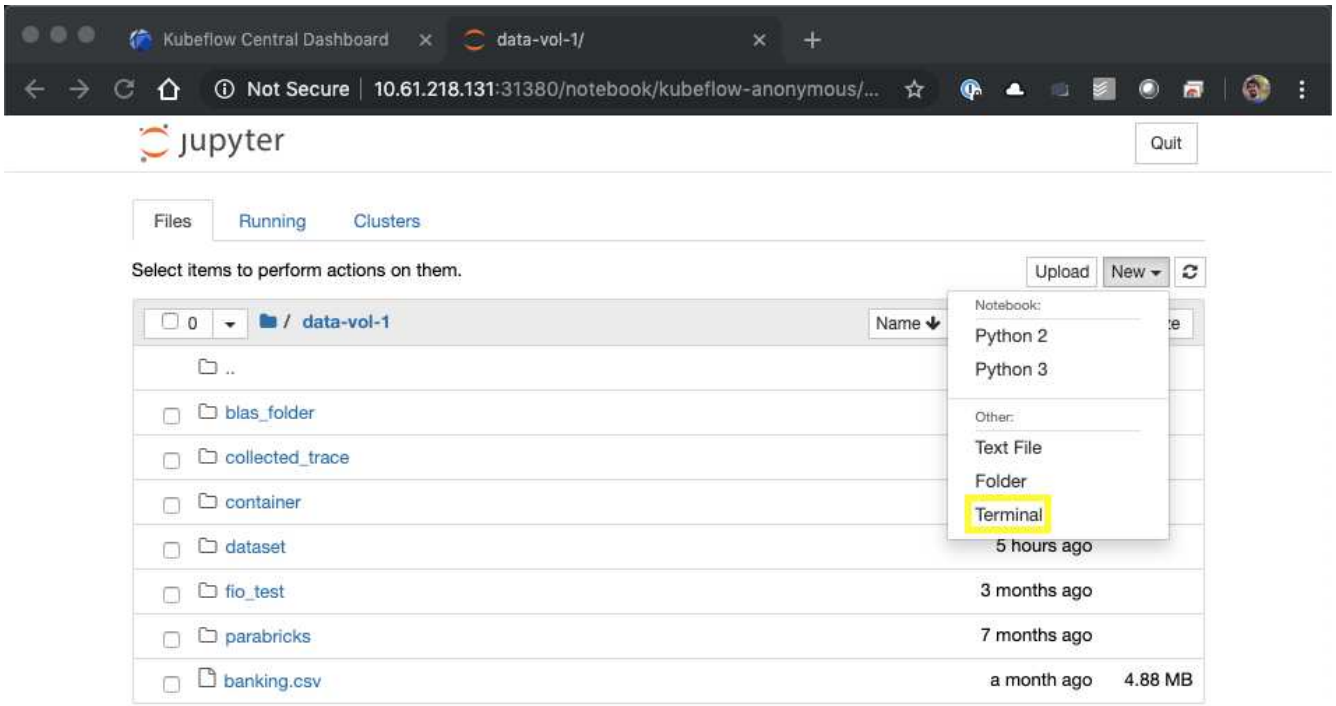
9. 按一下「連線」以連線至新的伺服器Web介面。
10. 確認已在伺服器上掛載步驟6中指定的資料集磁碟區。請注意、此磁碟區預設會掛載於預設工作區內。從使用者的觀點來看、這只是工作區內的另一個資料夾。使用者可能是資料科學家、而非基礎架構專家、因此不需要擁有任何儲存專業知識、就能使用此磁碟區。

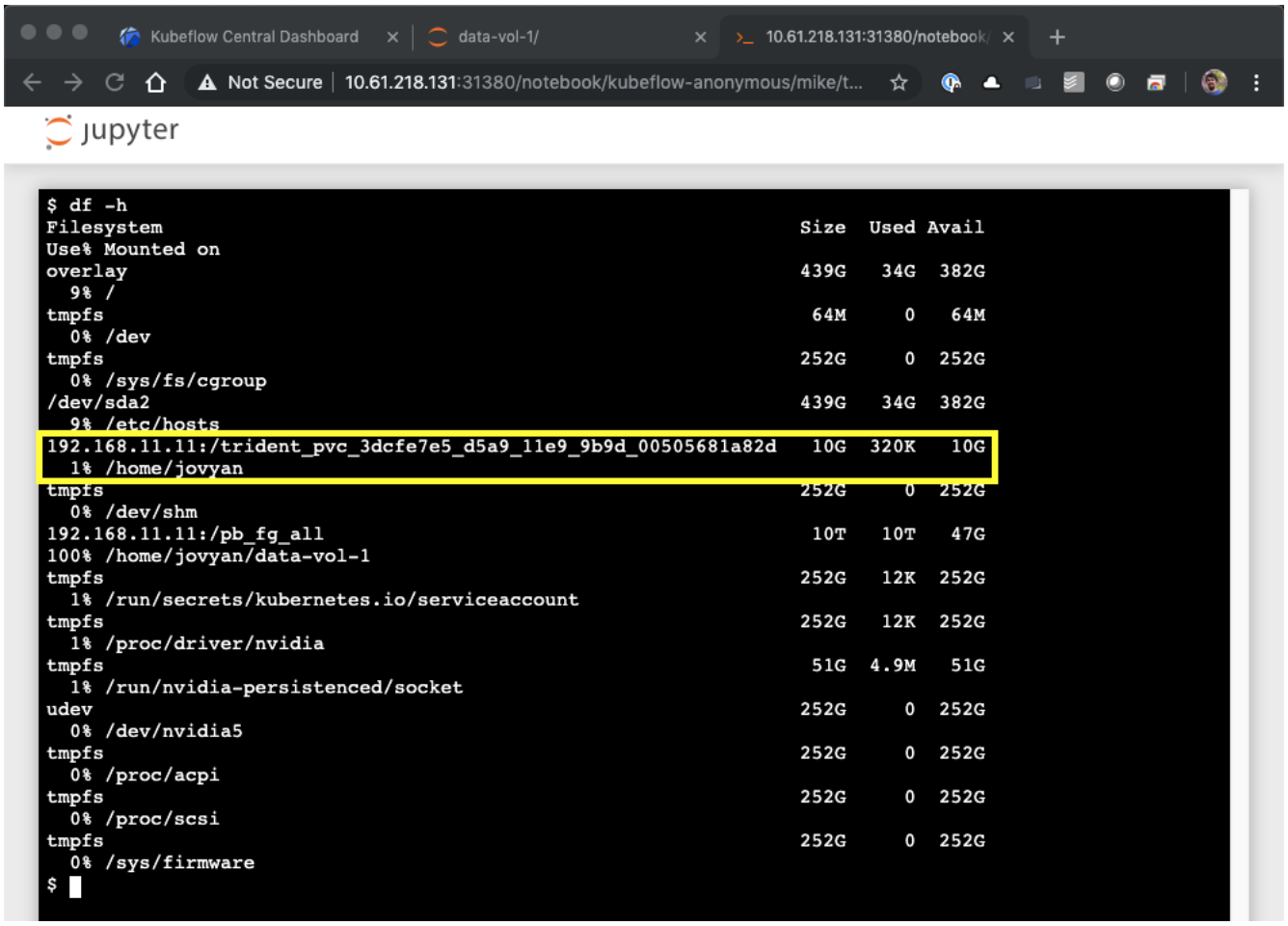




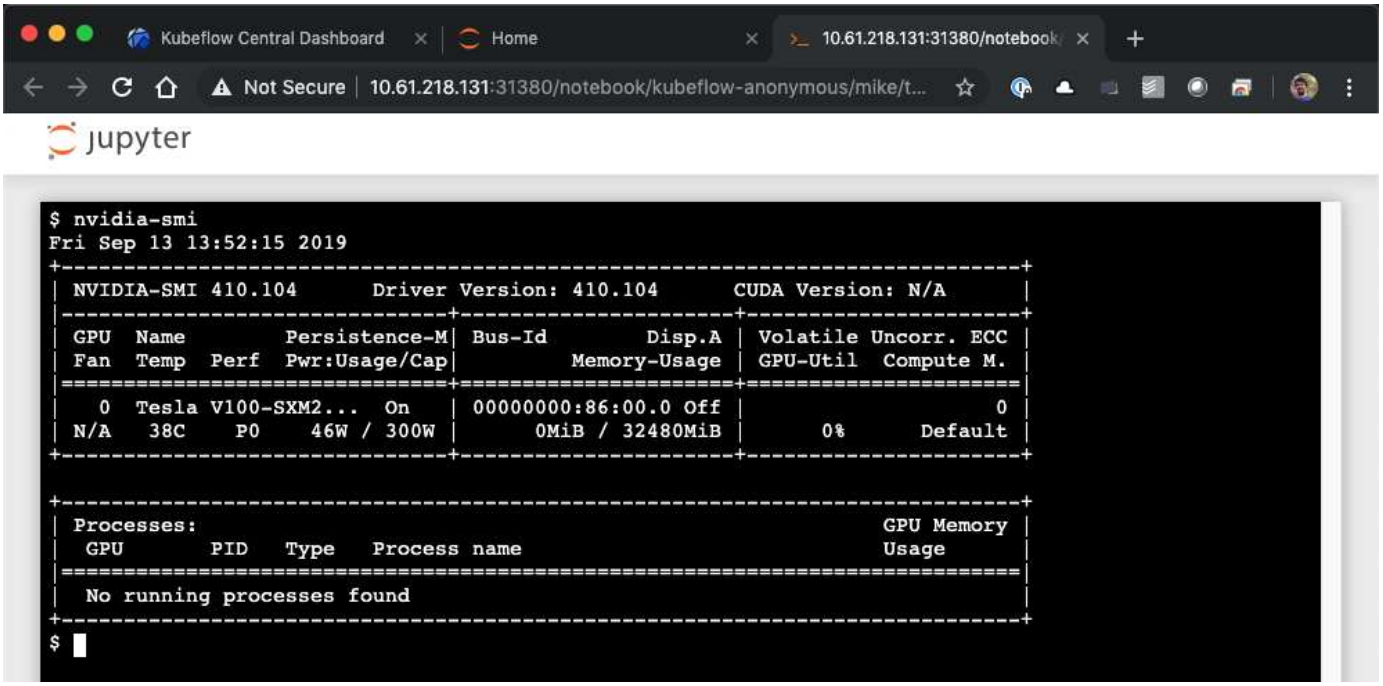
11. 開啟終端機、並假設步驟5要求新磁碟區、執行「df -h」、確認新的Trident資源配置持續磁碟區已掛載為預設工作區。

    預設工作區目錄是您第一次存取伺服器Web介面時所顯示的基礎目錄。因此、您使用Web介面建立的任何成品都會儲存在此Trident資源配置的持續Volume中。





12. 使用終端機執行「nvidia-smi」、確認已將正確數量的GPU分配給筆記型電腦伺服器。在下列範例中、已依照步驟7的要求、將一個GPU分配給筆記型電腦伺服器。



"下一步：筆記型電腦和管線範例。"

筆記型電腦和管線範例

- "適用於Kubernetes的NetApp Data科學工具套件" 可與Kubeflow搭配使用。搭配Kubeflow使用NetApp Data科學工具套件可提供下列效益：

  - 資料科學家可直接從Jupyter筆記型電腦執行進階NetApp資料管理作業。
  - 進階NetApp資料管理作業可透過Kubeflow Pipes架構整合至自動化工作流程中。

請參閱 "Kubeflow範例" NetApp Data科學工具套件GitHub儲存庫中的一節、詳細說明如何搭配Kubeflow使用此工具組。

"下一步：Apache Airflow部署。"

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.